introduce spark to ignite the air-fuel mixture. Spark plug consists of a central electrode, a ground electrode, an insulator. There is a small gap (0.5mm to 0.7mm) is available between the central electrode and ground electrode. When the high tension current from the ignition coil through a distributor is applied between these electrodes, spark is produced. This spark will ignite the air fuel mixture.

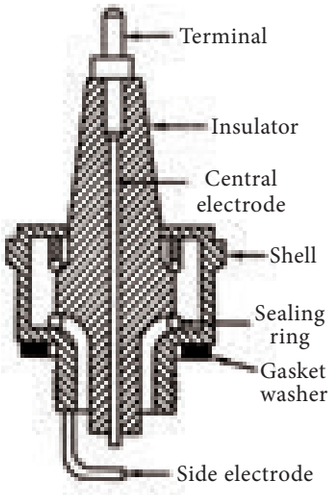


Figure 10.4 (b) Spark plug

10.4.1 Types of spark plug

1. Hot spark plug
2. Cold spark plug

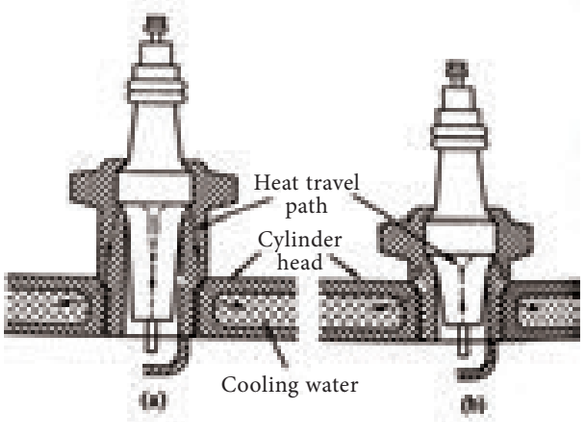


Figure 10.4.1 (a) Types of spark plug

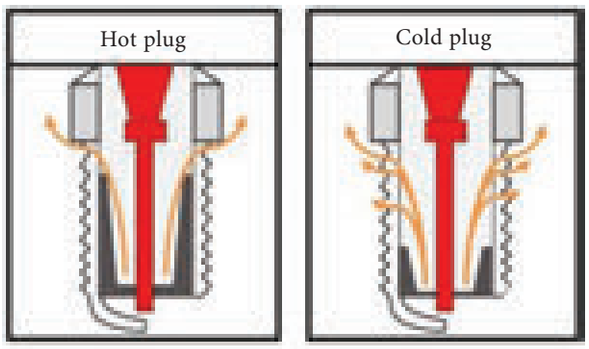


Figure 10.4.1 (b) Types of spark plug

Hot spark plug

In this type of spark plug, the length of the insulator is long and heat travel path is long. This spark plug is widely used in low-speed engine and engine operating in the cold climatic area.

Cold spark plug

In this, the length of the insulator is short and heat travel part is short. These spark plugs are widely used in high-speed engines and heavy loaded engines.

10.4.2 Ignition switch and key

The Ignition switch is used to connect the battery to the required circuits via a key. It will have three terminal viz., The battery (B), Ignition (Ig) and Starter (St). The terminal B is always connected to the battery. Ignition system, fuel system, lighting and other electrical accessories are connected in the Ignition terminal via a fuse. The solenoid switch of the starter motor is connected in Starter (St) terminal.

When the ignition switch is positioned in Ignition (Ig), then the electricity from the

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